day / date:

⇒ proof of Theorum 3

$a \equiv b \pmod{m}$

* By def of notation before theorum 2 : $m \mid a-b$

$$a-b = mk \qquad k \in \mathbb{Z}$$

$$\boxed{a = b + mK}$$

Theorum 4: let m be +ve integer., if $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$. Then:

$$a+c \equiv b+d \pmod{m}$$
$$ac = bd \pmod{m}$$

Ex: If $7 \equiv 2 \pmod{5}$, $11 \equiv 1 \pmod{5}$

$$7+11 = 2+1 \pmod{5} \rightarrow = 3 \pmod{5}$$
$$7 \times 11 = 2 \times 1 \pmod{5} \rightarrow = 2 \pmod{5}$$

Ex: 4.3 → 4.2 not included

LECTURE-13

Monday 6/10/25.

Primes & GCD:

DEF. An integer $p > 1$ is called prime if the only factor of p are 1 and p itself.
e.g: 2, 3, 5, 7 ...

Composite: The integer n is composite iff there is an integer 'a' such that $a \mid n$ and $1 < a < n$.

Writing a composite no. in product of prime no.
OR " " prime no. as prime no.

day / date:

⇒ Fundamental Theorum of Arithmetic (Prime Factorization)

Every integer greater than 1 can be written uniquely as a prime or as the product of two or more primes where factors are written in order of non-decreasing size

Example:

(a) 27. Write prime factorization:
* Start with least prime no. i.e: 2 to see if number (27) is divisible by it.
since: 2×27
$\therefore$: $3 \cdot 9 = 27$
* Since '9' isn't prime $\therefore$ $2 \times 9 \Longrightarrow (3 \cdot 3) = 9$

$$\therefore 27 = 3 \cdot (3 \cdot 3) \Rightarrow \boxed{3^3}$$

(b) 100.

$$100 = 2 \cdot 50$$
$$2 \cdot 2 \cdot 25$$
$$2 \cdot 2 \cdot 5 \cdot 5$$
$$= \boxed{2^2 \cdot 5^2}$$

shldn't be '$5^2 2^2$' since order matters

(c) $500 = 2^2 \cdot 5^3$

⇒ Find GCD using Prime Factorisation.

$$a = p_1^{a_1} \cdot p_2^{a_2} \cdot p_3^{a_3} \cdots p_n^{a_n}$$

$$b = p_1^{b_1} \cdot p_2^{b_2} \cdot p_3^{b_3} \cdots p_n^{b_n}$$

either a_1 or b_1

$$\gcd(a,b) = p_1^{\min(a_1, b_1)} \cdot p_2^{\min(a_2, b_2)} \cdots p_n^{\min(a_n, b_n)}$$

day / date:

Q: Prime factorisation ~~&~~ GCD of 36, 84.

Prime factorisation:

$36 = 2^2 \cdot 3^2 \; (7^0)$ — 7^0, not 7^1 as 7^0 is min

$84 = 2^2 \cdot 3 \cdot 7$

3^1, not 3^2 since 3^1 is minimum

$gcd(34, 84) = 2^2 \cdot 3 \cdot 7^0$
$= 4 \cdot 3 \cdot 1$
$= \boxed{12}$.

☐ Greatest Common Factor (GCD / HCF):

- Let a & b be integers (not both zero). The largest integer d such that $d \mid a$ & $d \mid b$ is called the gcd of a & b.
- That is, if $\exists\, e \neq d \in \mathbb{Z}^+$, s.t $e \mid a$ & $e \mid b$ then $e < d$.

☐ Least Common Multiple (LCM)

- The LCM of the +ve integer a & b is the smallest +ve integer m that is ~~divisible~~ divisible by both a & b.
- That is, if $\exists\, n \neq m$ & $a \mid n$ & $b \mid n$, then $m < n$

Formula:

$$a = p_1^{a_1} \cdot p_2^{a_2} \cdots\cdots p_n^{a_n}$$
$$b = p_1^{b_1} \cdot p_2^{b_2} \cdots\cdots p_n^{b_n}$$

$$LCM(a,b) = p_1^{\max(a_1,b_1)} \cdot p_2^{\max(a_2,b_2)} \cdots p_n^{\max(a_n,b_n)}$$

day / date:

Q: Use prime factorization to find GCD & LCM of:

(a) $a = 120$, $b = 500$

$120 = 2 \cdot 60 \to 2 \cdot 2 \cdot 30 \to 2 \cdot 2 \cdot 2 \cdot 15 \to \boxed{2^3 \cdot 3 \cdot 5}$

$500 = 2 \cdot 250 \to 2 \cdot 2 \cdot 125 \to 2 \cdot 2 \cdot 5 \cdot 5 \cdot 5 = \boxed{2^2 \cdot 5^2}$

$GCD(120, 500) = 2^2 \cdot 3^0 \cdot 5$
$= \boxed{20}$

since out of 2^2 & 2^3, 2^3 is max.

$LCM(120, 500) = 2^3 \cdot 3 \cdot 5^2$
$= \boxed{600}$

(b) $a = 17$, $b = 22$.

$17 = \boxed{17}$

$22 = \boxed{2 \cdot 11}$

$GCD = \cancel{\;} \cdot 2^0 \cdot 17^0 \cdot 11^0$
$= \boxed{1}$ (Relatively prime since GCD = 1)

$LCM = 2 \times 17 \times 11$
$\boxed{374}$

☐ Relatively Prime: The integer a & b are relatively prime if $gcd(a,b) = 1$

☐ Pairwise Relatively Prime: The integers $a_1, a_2, a_3 \ldots a_n$ are pairwise relatively prime if $gcd(a_i, a_j) = 1$, whenever $1 \leq i < j \leq n$.

day / date:

Q: Determine wheather integers are pairwise relatively prime or not:

(a) 10, 17, 21

$10 = 2 \cdot 5$, $17 = 17$, $21 = 3 \cdot 7$

$\gcd(10,17) = 2^0 \cdot 5^0 \cdot 17^0 = \boxed{1}$

$\gcd(17,21) = 17^0 \cdot 3^0 \cdot 7^0 = \boxed{1}$

$\gcd(10,21) = 2^0 \cdot 5^0 \cdot 3^0 \cdot 7^0 = \boxed{1}$

since all three gcd's = 1 ∴ it is pairwise relative prime.

(b) 10, 19, 24

$10 = 2 \cdot 5$, $19 = 19$, $24 = 2^3 \cdot 3$

$\gcd(10,19) = 1$

$\gcd(10,24) = 2$

different ∴ not pairwise relatively prime.

EUCLDEAN ALGORITHM : Successive use of division algo.

Lemma:

let $a = bq + r$, where $a, b, q, r \in \mathbb{Z}$; $a > b$

then $\boxed{\gcd(a,b) = \gcd(b,r)}$

Q: Find gcd using Eucldean Algo.

(a) $\gcd(91, 287)$ — $a > b$, 91 → b, 287 → a

$a = bq + r$

$287 = 91(3) + 14$

Division Algorithm from prev. lec.

$91 = 14(6) + 7$

$14 = 7(2) + 0$

$\gcd(91, 287) = \boxed{7}$

day / date:

b) $\gcd(277, 123)$ — a = 277, b = 123

$277 = 123(2) + 28$

$123 = 28(4) + 11$

$28 = 11(2) + 6$

$11 = 6(1) + 5$

$6 = 5(1) + 1$

$5 = 1(5) + 0$

for gcd, we'll not stop until the remainder is zero.

$\gcd(277, 123) = \boxed{1}$

Theorum 6: BEZONT'S THEORUM.

If a & b are +ve integers then there exist integer s & t such that gcd as linear combo. of a & b.

$$\boxed{\gcd(a,b) = sa + tb}$$

Bezont's Identity / Linear Combo

DEF: If a & b are +ve integers, then s & t such that $\gcd(a,b) = sa + tb$ are called "bezont's cofficient" of a & b.

Q: Express a & b as Linear Combo. of a & b.
Express $\gcd(a,b)$ as Linear Combo. of a & b.

(a) $a = 91$, $b = 287$

★ First perform Eucldean Algo.

$287 = 91(3) + 14$ ①

$91 = 14(6) + 7$ ②

jitnay Algos. hon gay utni dafa remainder ko substitute krna.

$14 = 7(2) + 0 \rightarrow \gcd(,) = \boxed{7}$

- $\gcd = sa + tb$

$7 = s(91) + 287t$

NOTE: In Eucldean Algo, we are successively applying division algo.

day / date:

* making '7' subject from eq ②

$7 = 91 - (6 \cdot 14)$ → value of 14 from eq ①.

$= 91 - 6(287 - 3 \cdot 91)$

$= 91 - 6 \cdot 287 + 18 \cdot 91$

$7 = 19(91) + (-6)(287)$

* 19 & −6 are Bezont's cofficients.

(b) $a = 252$, $b = 198$

$252 = 198(1) + 54$ → $198 = 54(3) + 36$

$54 = 36(1) + 18$

$36 = 18(2) + 0$

gcd = 18 ⟶ gcd = sa + tb

$18 = s(252) + t(198)$

$18 = 54 - 1 \cdot 36$

$= 54 - 1(198 - 3 \cdot 54)$

$= 4 \cdot 54 - 1 \cdot 198$

$= 4(1 \cdot 252 - 1 \cdot 198) - 1 \cdot 198$

$= 4 \cdot 252 - 4 \cdot 198 - 1 \cdot 198$

$18 = 4(252) + (-5)(198)$

* 4 & −5 are Bezont's cofficients.

Q: Express gcd(101, 4620) as a linear combo. of 101 & 4620

write Bezout's cofficients of a & b.

Euclidean: Division Algo.

$4620 = 101(45) + 75$	Values of remainders (for substitution)
$101 = 75(1) + 26$	$75 = 4620 - 45 \cdot 101$
$75 = 26(2) + 23$	$26 = 101 - 1 \cdot 75$
$26 = 23(1) + 3$	$23 = 75 - 2 \cdot 26$
	$3 = 26 - 1 \cdot 23$

day / date:

$23 = 3(7) + 2$	values of remainders.
$3 = 2(1) + 1$	$2 = 23 - 7 \cdot 3$
$2 = 1(2) + 0$	$1 = 3 - 1 \cdot 2$

∴ gcd(101, 4620) = 1

⇒ Back Substitution: (from 2nd last step)

gcd = $1 = 3 - 1 \cdot 2$

* Substitue value of '1' from 2nd last step. $(1 = 3 - 2 \cdot 1)$

$1 = 3 - 1 \cdot 2$

* substitute value of '2' $(2 = 23 - 7 \cdot 3)$

$1 = 3 - 1 \cdot (23 - 7 \cdot 3)$

$1 = 1 \cdot 3 - 1 \cdot 23 - 7 \cdot 3$

$= 8 \cdot 3 - 1 \cdot 23$

* substitute value of '3' $(3 = 26 - 1 \cdot 23)$

$= 8(26 - 1 \cdot 23) - 1 \cdot 23$

$= 8 \cdot 26 - 8 \cdot 23 - 1 \cdot 23$

$= 8 \cdot 26 - 9 \cdot 23$

* Substitude value of '23' $(23 = 75 - 2 \cdot 26)$

$8(26) - 9 \cdot (75 - 2 \cdot 26)$

$8 \cdot 26 - 9 \cdot 75 - 18 \cdot 26$

$26 \cdot 26 - 9 \cdot 75$

* substitude value of 26 $(26 = 101 - 1 \cdot 75)$

day / date:

$= 26(101 - 1\cdot 75) - 9\cdot 75$

$= 26 \cdot 101 - 26\cdot 75 - 9\cdot 75$

$= 26\cdot 101 - 35\cdot 75$ → value of 75

$= 26\cdot 101 - 35(4620 - 45\cdot 101)$

$= 26\cdot 101 - 35\cdot 4620 - 15.75\cdot 101$

$1 = \underset{s}{(1601)}\underset{a}{(101)} - \underset{t}{(35)}\underset{b}{(4620)}$

∴ 1601 & -35 are the Bezout's Cofficients.

LECTURE-14

Monday 13/10/25

$\boxed{\text{Ex: 4.4}}$: Solving Congruencies

⇒ Linear Congruence: A congruence of the form $ax \equiv b \pmod m$, where $m > 0 \in \mathbb{Z}^+$ and $a, b \in \mathbb{Z}$, 'x' is a variable.

- Solution of a linear Congruence are all integers x that satisfies the congruence.

Inverse: An integer $\boxed{\bar{a}}$ such that $\bar{a}a = 1 \pmod m$ is said to be the inverse of '$a \bmod m$'

Inverse By Inspection: Using inspection to find an inverse of '$a \bmod m$' is easy when m is small. To find the inverse, we look for a multiple of 'a' that exceeds multiple of 'm' by 1.

for smaller m (like: 7) so we can Hit & Trail.

day / date:

Eg: Find Inverse of $\overset{a}{3} \bmod \overset{m}{7}$

Hit & Trail.

By def. of Inverse → $\bar{a}3 \equiv 1 \pmod 7$

$7 \backslash \bar{a}\cdot 3 - 1$

* 7 should divide RHS fully with remainder 0.

as in $a \equiv b \bmod m$

$b \backslash a - b$

$7 \backslash 3\bar{a} - 1$

* Let $\bar{a} = 0, 1, 2, 3, 4, 5, 6.$

$\bar{a} = 0$

$7 \backslash 0\cdot 3 - 1 \rightarrow 7 \backslash -1$ ✗

$\bar{a} = 1$

$7 \backslash 1\cdot 3 - 1 \rightarrow 7 \backslash 2$ ✗

$\bar{a} = 2$

$7 \backslash 2\cdot 3 - 1 \rightarrow 7 \backslash 5$ ✗

⋮

$\bar{a} = 5$

$7 \backslash 5\cdot 3 - 1 \rightarrow 7 \backslash 14 = 0$ ✓

∴ answer → $\boxed{\bar{a} = 5}$ a inverse is 5

$\overset{a}{3}x \equiv \overset{b}{4} \bmod \overset{m}{7} \rightarrow ①$

⇒ To Solve the congruence we must have the Inverse of 3 mod 7

$\bar{a} = 5$

$a\bar{a} = 1 \pmod m$

* Multiply ① by $\bar{a}$ (on both RHS & LHS)

$(\overset{\bar{a}}{5}\cdot \overset{a}{3})x = \overset{\bar{a}}{5}\cdot \overset{b}{4} \pmod{\overset{m}{7}}$

* Since $5\cdot 3 \equiv 1 \pmod 7$ ∴ $5\cdot 3 = 1$

$1x \equiv 20 \bmod (7)$

$$\begin{array}{r} 2 \\ 7\overline{)20} \\ -14 \\ \hline 6 \end{array}$$

$\boxed{x \equiv 6 \pmod 7}$

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day / date:

$\boxed{\text{RULE}}$ was : $a \equiv b \pmod m$
means $m \mid a-b$

$\therefore$

$x \equiv 6 \pmod 7$

$7 \mid x-6$

$x-6 = 7t$, $t \in \mathbb{Z}$

$\boxed{x = 7t+6}$ answer.

Q: Solve linear congruence by Inspection method. $2x \equiv 7 \bmod 17$

$\bar{a}2 = 1 \bmod 17$

$\bar{a} = 9$

$9 \cdot 2 \equiv 1 \pmod{17}$

* multiply ① by $\bar{a} = 9$

$9 \cdot 2\, x \equiv 9 \cdot 7 \pmod{17}$ → since $9 \cdot 2 \equiv (1 \bmod 17) = 1$

$x \equiv 63 \bmod 17$

$x \equiv 12 \bmod 17$

$$17 \overline{)63}^{\,3} \quad -51 \quad 12$$

$17 \mid x-12$

$x-12 = 17t$ $t \in \mathbb{Z}$.

$\boxed{x = 17t - 12}$

→ Can't do Hit & trail. A bigger no. like 4620 occurs.

Q: Find an Inverse of of $\underset{a}{101}$ mod $\underset{m}{4620}$

1) gcd
2) Eucledean
3) Bezout's

from previous to previous page.

$\gcd(101, 4620) = sa + tb$

$1 = 1601(101) + \underbrace{(-35)(4620)}_{0}$

$\therefore 1 \pmod{4620} = 1601(101)$

* cofficient of a is $\bar{a}$ which is inverse

$\therefore \bar{a} = \boxed{1601}$

Solve system of Congruencies to find unique sol.

day / date:

The Chinese Remainder Theorum :

Let $m_1, m_2 \dots m_n$ be pairwise relatively prime +ve integer greater than one and $a_1, a_2, a_3 \dots a_n$ be arbitary integers

Then the system

$$x \equiv a_1 \pmod{m_1}$$
$$x \equiv a_2 \pmod{m_2}$$
$$\vdots$$
$$x = a_n \pmod{m_n}$$

has a unique sol. mod m

where $m = m_1 m_2 m_3 \dots m_n$

~~Q: Find $m_1, m_2, m_3 \dots m_n$~~
~~Take $M_1 = \frac{m}{m_1}$, $M_2 = \frac{m}{m_2}$ ... $M_n = \frac{m}{m_n}$~~

Q: System of Sol. is given we need to find a unique sol.

$\boxed{\text{Step 1}}$: Find $m = m_1, m_2 \dots m_n$

Take : $M_1 = \frac{m}{m_1}$, $M_2 = \frac{m}{m_2}$ ... $M_n = \frac{m}{m_n}$

$\boxed{\text{Step 2}}$: Find Inverses

$M_1 \cdot y_1 \equiv 1 \pmod{m_1}$ $\quad M_1 \cdot y_1 = M \cdot \overline{M}$ (inverse of M)

$M_2 \cdot y_2 \equiv 1 \pmod{m_2}$

$\vdots$

$M_n \cdot y_n \equiv 1 \pmod{m_n}$

$\boxed{\text{Step 3}}$: Solve

$$x = (a_1 M_1 y_1 + a_2 M_2 y_2 + \dots a_n M_n y_n) \pmod m$$

day / date:

Q: Solve . $x \equiv 2 \bmod 3$

$x \equiv 3 \bmod 5$

$x \equiv 2 \bmod 7$

By Chinese Remainder Theorum.

1. $m = 3 \times 5 \times 7$

   $m = 105$

2. $M_1 = \frac{105}{3}$, $M_2 = \frac{105}{5}$, $M_3 = \frac{105}{7}$

   $M_1 = 35$, $M_2 = 21$, $M_3 = 15$

3. $M_1 \cdot \overline{M_1} = 1 \bmod 3$

   $35 \cdot \overline{M_1} = 1 \pmod 3$

   $3 \backslash 35(2) = 1$ Hit & trail from prev. class.

   $\therefore \boxed{\overline{M_1} = 2}$

   $M_2 \cdot \overline{M_2} = 1 \pmod 5$

   $21 \cdot 1 = 1 \pmod 5$

   $\boxed{\overline{M_2} = 1}$

   $M_3 \cdot \overline{M_3} = 1 \bmod 7$

   $15 \cdot 1 = 1 \bmod 7$

   $\boxed{\overline{M_3} = 1}$

4. $x \equiv (2)(35)(2) + (3)(21)(1) + (2)(15)(1) \pmod{105}$

   $x \equiv 233 \pmod{105}$

   $\boxed{x = 23}$ OR $x \equiv 23 \pmod{105}$ OR

   $x = 105t + 23.$

day / date:

Q. FERMAT's LITTLE Theorum:

If 'p' is prime and 'a' is an integer not divisible by 'p', then:

$$a^{p-1} \equiv 1 \pmod p$$

Furthermore, for every integer we have:

$$a^p \equiv a \pmod p$$

Q: Find $7^{222} \bmod 11$

$a = 7$, $p = 11$

$7^{11-1} \equiv 1 \bmod (11)$

$7^{10} \equiv 1 \pmod{11} \rightarrow$ A

• Take 10 till 222 using division algorithm.

$222 = 22(10) + 2$

$a = d\,q + r$

$7^{222} \equiv 7^{10(22)+2} \pmod{11}$

$= (7^{10})^{22} \cdot 7^2 \bmod 11$

* According to A, $(7^{10})^{22} = 1$ $\therefore$

$7^{222} = 1 \cdot 49 \bmod 11$

$\boxed{= 5}$

Q (a) USE FLT to compute

$x \equiv 5^{2003} \mod 7$,

$x \equiv 5^{2003} \mod 11$,

$x \equiv 5^{200} \mod 13$.

(b) Use results & Chinese remainder theorem to find:

$5^{2003} \pmod{1001}$

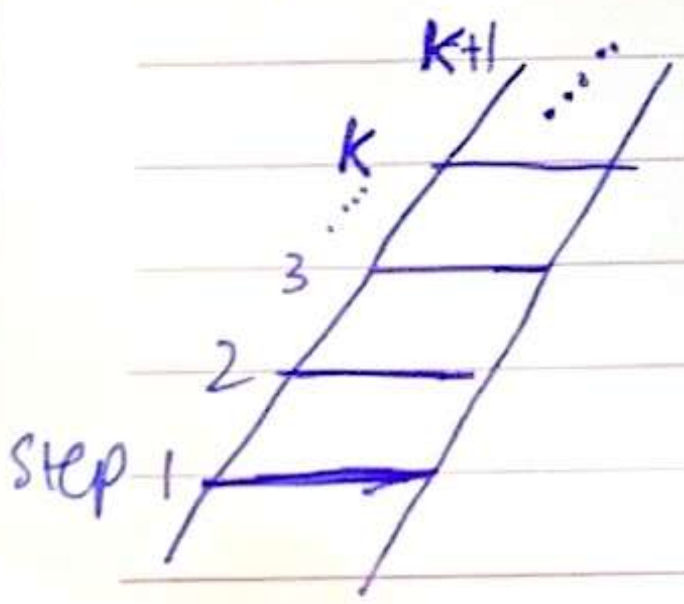
LECTURE-15 | Wednesday 15/10/25

5.1: Mathematical Induction.

To prove P(n) is true for all +ve integer n, where P(n) is a propositional function (mathematical statement), we complete two steps:

1) **Basis Step**: we verify that P(1) (initial value) is true. (Proposition is true on initial value)

2) **Inductive step**: we show that the conditional statement $P(K) \to P(K+1)$, $\forall K \in \mathbb{Z}^+$.
If P(K) is true; P(K+1) is true as well.



* If on an infinity ladder we are on k^{th} step; we can move on $(k+1)^{th}$ step.

⇒ Rule of Inference:

$$[P(1^{st}) \wedge \forall K (P(K) \to P(K+1))] \to \forall n\, P(n)$$

⇒ TYPES OF QUESTION:

- Summation (sum of arithmetic etc)
- Inequality
- Divisibility

Q: Show that if 'n' is a +ve integer, then

Prove $1+2+3+\dots n = \frac{n(n+1)}{2}$, $n \geq 1$

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}.$$

by mathematical induc.

Initial values: n is +ve → 1
n is non negative → 0.

Step 1: To show $P(1^{st})$ is true.

n = 1

$$1 = \frac{1(1+1)}{2}$$

∴ P(1) is true.

Step 2: Assume that proposition is true for k values.
$K \geq 1$ (K should start from the same value as 'n').

$$1+2+3 \cdots + K = \frac{K(K+1)}{2}$$ (Inductive hypothesis).

day / date:

To prove "K+1" is also true; replace 'K' in previous step with K+1:

$$\underbrace{1+2+3 \cdots +(K+1)}_{LHS} = \underbrace{\frac{K+1\,((K+1)+1)}{2}}_{RHS} \longrightarrow A$$

LHS → $\underbrace{1+2+3 \cdots +k}_{\frac{k(k+1)}{2}\text{ : Inductive hypothesis}} + (K+1)$. (expand a bit to use Inductive hypothesis).

$$\therefore \frac{k(k+1)}{2} + (k+1) = \frac{k(k+1) + 2(k+1)}{2}$$

RHS → $\frac{(K+1)(K+2)}{2}$ → since this is equal to RHS of A ∴ proved.

∴ This implies that $P(K+1)$ is true. $P(n)$ is true for all $n \geq 1$.

Q: Use M.I to show

$1 + 2 + 2^2 + \cdots 2^n = 2^{n+1} - 1$; for all non-negative 'n'. "n=0" is first value.

n=0

1. $2^{0+1} - 1 = 1$

   $2^1 - 1 = 1$

   $1 = 1$; ∴ $P(1^{st})$ is true. ($P(0)$ is true).

2. Let $P(k)$ is true

   I.H → $1+2+2^2+\cdots 2^k = 2^{k+1} - 1$

To prove $(K+1)$ is also true, $K \geq 0$

$$1+2+2^2 \cdots 2^{k+1} = 2^{(k+1)+1} - 1 \longrightarrow A$$

day / date:

$$\underbrace{1+2+3 \cdots + 2^k}_{\cancel{2} 2^{k+1} - 1 \text{ (by I.H)}} + 2^{(k+1)}$$

$$\therefore (2^{k+1} - 1) + 2^{k+1}$$

$$= 2(2^{k+1}) - 1 = \boxed{2^{k+2} - 1} \rightarrow \text{RHS.}$$

∴ $P(n)$ is true for all $n \geq 0$.

Q: Conjecture a formula for the sum of first n +ve odd integers. Then prove the formula using M.I.

To form a formula; we must take few values of n. As $n \geq 1$:

Take $n = 1, 2, 3, 4$.

n=1	1 = 1	1^2	Sum of first one odd
n=2	1 + 3 = 4	2^2	" " " two odds
n=3	1+3+5 = 9	3^2	" " " three odds
n=4	1+3+5+7 = 16	4^2	" " " four odds.

FORMULA: $n \geq 1$

$$1+3+5 \cdots + 2n - 1 = n^2$$

★ Now use same method 1 & 2 to prove using MI.

n = 1

1. $P(1) \rightarrow 1 = 1^2$

   $1 = 1$

   ∴ $P(1)$ is true.

day / date:

② I·H → $1+3+5+ \dots 2k+1 = k^2$

* To prove $K+1$ is true : $K \geq 1$.

LHS RHS.

$1+3+5 \dots 2(k+1)+1 = (k+1)^2 = k^2+2k+1$

LHS → $\underbrace{1+3+5 \dots (2k+1)}_{k^2 \text{ (by I H)}} + [(2k+1)+1]$

$k^2 + 2K + 2$.

since LHS = RHS ∴ P(n) is true for all $n \geq 1$.

Q: Prove that $n < 2^n$, for all $n \geq 1$.

$n = 1$

① $P(1) \rightarrow 1 < 2^{(1)}$

$1 < 2$ ∴ P(1) is true.

② I·H → $\boxed{K < 2^K}$, $K \geq 1$

* To prove $K+1$ is true :

$K+1 < 2^{K+1}$, $K \geq 1$

$< 2^K$ (by I·H).

LHS = $K + 1$

LHS = $K + 1 < 2K + 1$

day / date:

★ Since we've to convert "$2k+1$" to "2^{k+1}" ∴ we need to remove '1' from here and for that we'll represent 1 in terms of 2^k.

$2^k + 1$

as $1 < 2^k$ ∴ 1 represented in 2^k so we can work only with 2^k term & not $2^k + 1$.

$2^k + 1 < 2^k + 2^k$

$< 2^k (1+1)$ // 2^k common.

$< 2^k (2)^1$

$< \boxed{2^{k+1}} \rightarrow$ RHS.

∴ P(n) is true for all $n \geq 1$, as $k+1 < 2^{k+1}$

Q: Prove $2^n < n!$, for all $n \geq 4$.

① $n=4$

$P(4) \rightarrow 2^4 < 4!$

$16 < 24$. ∴ P(4) is true.

② I·H → $\boxed{2^k < k!}$

* For proving $k+1$; $\forall k \geq 4$:

LHS RHS.

$2^{k+1} < (k+1)!$

$2^{k+1} = 2 \cdot 2^k$

I·H → $2^k < k!$ (replace 2^k with $k!$)

∴ $2 \cdot 2^k < 2 \cdot k! \rightarrow$ ⊗

* we need to convert "$2 \cdot k!$" into $(k+1)!$ ∴ we need to remove '2'.

$\boxed{2 < k+1} \rightarrow$ plug in ⊗.

$2 \cdot 2^k < 2 \cdot k! < (k+1)k! \rightarrow 2 \cdot 2^k < (k+1)k!$

this is $(k+1)!$ ∴ RHS proved.

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Q: USE MI to prove :

(a) $n^3 - n$ is divisble by 3 ; $n \geq 1$

① $n = 1$

$1^3 - 1 = 0$

0 is divisble by 3 $\therefore$ P(1) is true

② I.H $\rightarrow$ $K^3 - K$ is divisible by 3

* To prove $(K+1)^3 - (K+1)$ is divisible by 3.

$(K^3 + 3K^2 + 3K + 1) - (K+1)$

$K^3 + 3K^2 + 3K + \cancel{1} - K - \cancel{1}$

$= (K^3 - K) + 3(K^2 + K) \longrightarrow A$

* As ① first term is divisible by 3 due to I.H and ② second term is divisible by 3 since its a multiple of '3' $\therefore$ we can say.

$a \backslash 3$ & $b \backslash 3$.

RULE : ~~If a\3 & b\3 ∴~~

If $3 \backslash a$ & $3 \backslash b$ $\therefore$ $3 \backslash a+b$.
and "a+b" is our eq A $\therefore$ A is divible by 3.

P(n) is true for $n \geq 1$

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day / date:

Q: $4^{n+1} + 5^{2n-1}$ is divisble by 21, $n \geq 1$

① $n = 1$

$P(1) \rightarrow 4^{1+1} + 5^{2(1)-1}$

$16 + 5 = 21$

$21 \backslash 21$ $\therefore$ P(1) is true.

② I.H $\rightarrow$ Assume $4^{k+1} + 5^{2k-1}$ is divisble by 21

* To prove $4^{(k+1)+1} + 5^{2(k+1)-1}$ is divisble by 21

$2(k+1) - 1 = 2k + 2 - 1$

$5^{2(k+1)-1} = 5^2 \cdot 5^{2k-1}$

$= 4^1 \cdot 4^{k+1} + 5^2 \cdot 5^{2k-1}$

$= 4 \cdot 4^{k+1} + 25 \cdot 5^{2k-1}$

$= 4 \cdot 4^{k+1} + (4 + 21) \cdot 5^{2k-1}$

$= 4 \cdot 4^{k+1} + 4 \cdot 5^{2k-1} \cdot 21 \cdot 5^{2k-1}$

$= \underbrace{4(4^{k+1} + 5^{2k-1})}_{a} + \underbrace{21 \cdot 5^{2k-1}}_{b}$

* Since 'a' is divible by 21 through I.H & 'b' is divisble by 21 as its a multiple of 21.

$\therefore$ as $21 \backslash a$ & $21 \backslash b$ $\therefore$ $21 \backslash a+b$.

P(n) is true for all $n \geq 1$.

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LECTURE-16 : Monday 20/10/25

Ex: 5.2

STRONG INDUCTION:

To prove P(n) is true for all positive integer 'n'. we complete two steps:

Basic step: Verify that the proposition is true on initial value(s).

Inductive step: we assume

$\forall j \ P(j)$ holds, where $1 \leq j \leq K$

That is:

$P(1) \wedge P(2) \wedge \dots \wedge P(K)$ is true for $K \geq 1$

we prove $P(k+1)$ is true.

INFINITE LADDER:

Suppose:

① we can reach the first and 2^{nd} rung of an infinite ladder (same ladder as math induction)

② If we can reach a (particular) rung then we can reach two steps higher.

Math Induction: $P(1) \wedge \forall k (P(K) \rightarrow P(k+1))$: Can move 1 step ahead only.

Strong Induction $(P(1) \wedge P(2) \dots \wedge P(K)) \rightarrow P(K+1)$: Can move 2 steps ahead

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NOTE: we still have to prove P(k+1). In math induction; we have to do it with P(k) but in strong Induction; we [redacted] can do it with P(K-1).

M.I	S.I
Basic: P(1) is true as we can reach 1^{st} step of ladder	Basic: P(1) & P(2) is true as we can reach first two steps.
Induction: Assume P(k) is true we can reach P(k+1) step to (k+1) is true.	Induction: Assume IH: $\forall j \ P(j)$ holds (where $1 \leq j \leq K$ and $K \geq 2$) since first two alr proved in basic step. • Using IH, P(k-1) is true since $K-1 < K$ • Adding 2 in K-1; we have K+1 ∴ P(K+1) is also true. P(n) is true for $n \geq 1$

Q: If 'n' is an integer greater than 1, then ~~se~~ 'n' can be written as the product of primes. → GPT like 10 year old

P(n) can be written as product of primes, $n > 1$.

Basic: $n \geq 2$

P(2) is true, since '2' is prime

Inductive: Assume $\forall j (2 \leq j \leq K \quad K \geq 2)$ each 'k' can be written as product of prime.

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To prove $P(k+1)$:

$\boxed{\text{Case 1}}$: If $k+1$ is prime; then $P(k+1)$ is true.

$\boxed{\text{Case 2}}$: if $k+1$ is composite; then

$$k+1 = a \cdot b \rightarrow (A)$$

~~$a \neq 1$ OR $b \neq 1$~~
~~$a \neq k+1$ OR $b \neq k+1$~~ → divisor cannot be greater than 'k'.

$\therefore 2 \leq a \leq k$, $2 \leq b \leq k$.

★ Using IH: $P(a)$ & $P(b)$ is true.
(a & b can be written as product of primes).

★ Prime factorization of a & b
$a = x \cdot y \cdot z$
$b = p \cdot q$
$a \cdot b = x \cdot y \cdot z \cdot p \cdot q$ (product of primes)

★ since $a \cdot b = k+1$ by (A)
$\therefore k+1 = xyzpq$ (prod. of primes) ✓

$\boxed{\text{Method}}$: $P(2)$ true
IH: $2 \leq n \leq k$; where n can be written as product of prime
$P(k+1)$: $\boxed{\text{Case 1}}$: if $(k+1)$ is prime so sorted.
$\boxed{\text{Case 2}}$: if $(k+1)$ not prime then it means $k+1 = a \cdot b$ where $a, b < k$

→ since a & b dd be written as product of prime $\therefore$ k+1 dd also be written as prod. of prime as $k+1 = a \cdot b$ (A)

★ As in IH; any no. less than k can be written as product of prime $\therefore$ a & b can be written as product of primes too.

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day / date:

Q: Prove that every amount of postage of 12-cents or more can be formed using 4-cents and 5-cents stamps.

P(n): n-cents postage can be formed using 4-cent and 5-cent typed stamps $n \geq 12$.

Basic: $n \geq 12$

$n = 12$	4 (3)	+ 5 (0)
$n = 13$	4 (2)	+ 5 (1)
$n = 14$	4 (1)	+ 5 (2)
$n = 15$	4 (0)	+ 5 (3)

Did this to find all combos of 4 & 5. That is: 4 & 5, 4 only, 5 only.

explain each step in words
12 cent stamps can be formed using three 4-cent stamps.

Inductive: Let $\forall j$ $P(j)$ holds (where $12 \leq j \leq k$ and $k \geq 15$) → IH
since we alr prove till $n=15$.

- To prove $(k+1)$ type cents stamps can be formed using 4 cents & 5 cents stamp
- Taking, $k-3$ from IH and $k-3 < k$:
  - if we add a 4 cent stamp in $(k-3)$ cents stamp we can form $k+1$ cents stamp.
  - $(k-3)+4 = \boxed{k+1}$ ✓

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day / date:

LECTURE 17

Wednesday 22/10/25

CHAP # 6.1

COUNTING

① Product RULE | ② SUM | ③ SUBTRACTION RULE | ④ DIVISIBILITY ×

① (Occurs when a procedure can be broken down into a sequence of task)
There are n_1 ways to do 1^{st} task, n_2 ways to do 2^{nd} task ..... n_m ways to do 'm' tasks.
Possible ways = $n_1 . n_2 ..... n_m$

② If a task can be done either in one of n_1 ways or one of n_2 ways, where none of the set of n_1 ways is same as n_2 ways.
possible ways = $n_1 + n_2$.

Ex: The chairs of an auditorium are to be labelled with two character, a letter followed by a digit. How many ways the chairs can be labelled?

Product Rule.

$\underset{1}{\underline{\text{letter}}}$ $\underset{2}{\underline{\text{digit}}}$

Possibilities of 1: (letter) = 26
" " 2: (digit) = 10

Total Possibilities = 26 × 10
= $\boxed{260}$

day / date:

Q: Find the no. of ways that an organization consisting 15 member can elect a president, a VP and treasury. (Assuming no person gets more than 1 position).

$\underset{1}{\underline{15}}$ $\underset{2}{\underline{14}}$ $\underset{3}{\underline{13}}$

1^{st} Position = 15
2^{nd} " = 14 (15 − 1)
3^{rd} " = 13 (14 − 1)

Total possibilities = 15 × 14 × 13
= $\boxed{2730}$.

Q: There're four bus lines b/w A & B.
" three " " " B & C

Find the no. of ways a person can travel:

(a) By bus from A to C by way of B

$A \xrightarrow{4} B \xrightarrow{3} C$ ∴ 4 × 3 = 12 ways.

(b) Round trip from A to C by way of B.

$A \xrightarrow[4]{} B \xrightarrow[3]{} C \xrightarrow[4]{} B \xrightarrow[3]{} A$

4 × 3 × 4 × 3 = $\boxed{144}$

(c) Round trip from A to C by way of B.
A person does not uses bus line more than once

$A \xrightarrow{4} B \xrightarrow{3} C \xrightarrow{2} B \xrightarrow{3} A$

4 × 3 × 2 × 3 = $\boxed{72}$.

Q: A student can choose a project from the 1st list is 23 ways, from 2nd list in 19 ways. How many ways a student can choose his project. Sum Rule.

Since he has to choose one project from any two list's therefore possibilities would be added as list's are different entities.

$\therefore$ $23 + 19 = \boxed{42}$ ways

$\Rightarrow$ Combining Sum / Prod:

Ex: Suppose the statement label in a programming lang. can be either a single letter or a letter followed by a digit. Find no. of possible label.

Single letter - ways $= \boxed{26}$

Double - ways $= \underline{26} \times \underline{10} = \boxed{260}$

Total ways $= 260 + 26$

$= \boxed{286}$.

Q: Computer access codeword consists of one to three letters of Eng. alphabets with repitation not allowed. How many diff. codewords are possible.

length 1 : $\underline{26}$ $= 26$

length 2 : $\underline{26} \times \underline{25}$ $= 650$

length 3 : $\underline{26} \times \underline{25} \times \underline{24}$ $= 15,600$

Total ways $= 26 + 650 + 15,600$

$= \boxed{16,276.}$

Q: Phone no. Generation.

X denotes values 0-9 (10)

N denotes values 2-9 (8)

Y denotes values 0-1 (2)

(a) Possibilities for : NYX - NNX - XXXX.

way $= 8^3 \times 2^1 \times 10^6$

(b) " " : NXX - NXX - XXXX

ways $= 10^6 * 8^2$

(3) SUBTRACTION RULE:

If a task can be done either in one of n_1 ways or in one of n_2 ways, then the no. of ways to do the task that are common to the two diff. ways is $n_1 + n_2$

Q: How many bit strings of length eight start either with a '1' bit or ends with the two bits '00'?

1 fixed, 2 ways to place bit (0/1)

starts with '1' bit $= \underline{1}\ \underline{2}\ \underline{2}\ \underline{2}\ \underline{2}\ \underline{2}\ \underline{2}\ \underline{2} = \boxed{2^7}$

ends with '00' bit $= \underline{2}\ \underline{2}\ \underline{2}\ \underline{2}\ \underline{2}\ \underline{2}\ \underline{0}\ \underline{0} = \boxed{2^6}$

starts with bit '1' & ends with '00' $= \underline{1}\ \underline{2}\ \underline{2}\ \underline{2}\ \underline{2}\ \underline{2}\ \underline{0}\ \underline{0}$

$= \boxed{2^5}$

Total ways $= 2^7 + 2^6 - 2^5$

$= \boxed{160}$.

day / date:

SET NOTATION:

SUM: $|(A_1 \cup A_2 \cup \dots A_n)| = |A_1| + |A_2| \dots |A_n|$.

PROD: $|A_1 \times A_2 \times \cup A_n| = |A_1| \cdot |A_2| \dots |A_n|$

SUBSTRACTION: $|A \cup B| = |A| + |B| - |A \cap B|$

Ex: 6.2 . <u>PIGEON-HOLE PRINCIPLE</u>:

If k is a +ve integer and $k+1$ objects are placed into 'K' boxes, then at least one box contains two or more objects. 6 boxes 7 peigon.

Generalized: If N objects are placed into 'K' boxes then there is atleast one box containing: $\lceil \frac{N}{k} \rceil$ objects.

To Find N:

$$N = K\left(\left\lceil \frac{N}{k} \right\rceil - 1\right) + 1$$

Q: What is the min. no of people I need to ensure that two people share same B'day.

K

- Days in year = 366 (leap) : worst case
- No. of people = $\boxed{367}$ (366+1) → min. no. of ppl.

* we did K+1 (i.e 366+1) so we can overflow people over the no. of days so there is a mutual b'day.

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Q: What is the min. no. of words I need to use (in eng lang.) to ensure that two words starts with the same alphabets.

No. of alphabets = 26 (k)

ways = 26+1 = $\boxed{27}$.

Q: Among 50 people, what is the min. no. of people that must be born on the same month?

No. of months = 12 (k)

$= \left\lceil \frac{50}{12} \right\rceil = \lceil 4.25 \rceil = \boxed{5}$ → ceiling func. for integer (not decimal)

Q: 38 diff time periods during which class of university can be scheduled. If there are 677 diff. courses, How many diff. rooms will be needed?

$\left\lceil \frac{677}{38 (k)} \right\rceil = \lceil 17.81 \rceil = \boxed{18}$ classes

Q: How many students in the class must there be to ensure that 3 students get the same grade (A, B, C, D or E)?

Grades = 5 = K

$\left\lceil \frac{N}{k} \right\rceil \rightarrow \left\lceil \frac{N}{5} \right\rceil = 3$, $N = ?$

Formula:

$N = \left(k\left\lceil \frac{N}{k} \right\rceil - 1\right) + 1 \longrightarrow N = 5(3-1) + 1$
$= \boxed{11}$.

* 11 students should be there in class to overflow the mark of atleast one grade = 3 students.

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LECTURE - 18

Monday
27/10/25.

Ex: 6.3 : Permutation & Combination

Intro: Many counting problems can be solved by finding the:

- no. of ways to arrange a specific no. of distinct elements of a set of particular size
- no. of ways to select a particular no. of elements from a set of particular size (unordered).

order matter

Permutation: Arrangements, sitting in a chair, standing in a row, word formatting, digit formatting.

Formula = $\boxed{P(n,r) \quad {}^nP_r} = \dfrac{n!}{(n-r)!}$

order doesn't matter

Combination: Selection, Collection, group makings, committee making

Formula = $C(n, r)$, ${}^nC_r$, $\binom{n}{r}$ → Binomial Coefficient

$\dfrac{n!}{(n-r)!\,r!}$

Q: 13 people on softball team show up for a game

(a) How many ways are there to choose 10 players to take the feild?

${}^nC_r \rightarrow {}^{13}C_{10} = \dfrac{13!}{(13-10)!\,10!} = \dfrac{13 \times 12 \times 11 \times \cancel{10!}}{3! \times \cancel{10!}} = \dfrac{13 \times \cancel{12}^{2} \times 11}{\cancel{6}} = \boxed{286}$

order matters.

(b) How many ways are there to assign the 10 positions by selecting players from the 13 players who show up.

$^{13}P_{10} = \frac{13!}{10!} = 13 \times 11 \times 12 = \boxed{1,716}$

Q2: How many ways are there to select a 1st prize winner, a 2nd prize winner & 3rd prize winner from 100 different people?

$^{100}P_3$ as order of 1st, 2nd, 3rd matters.

Q3 (a) How many poker hands of 5 cards can be selected from a standard deck of 52 cards.

$^{52}C_5$

(b) Also how many ways are there to select 47 cards from a deck of 52 cards?

$^{52}C_{47}$ as order still doesnot matter.

Q4: A coin is flipped 8 times where each flip comes up either heads or tails? How many possible ways:

(a) are there in total.

$n^r \rightarrow 2^n = 2^8 = \boxed{256}$

This is also permutation but here repetition is allowed:

$2\times2\times2\times2\times2\times2\times2\times2 = 2^8$

★Factorial is also a type of rep not allowed permutati

(b) contains exactly 3 Heads?

$^8C_3 = \frac{8!}{(8-3)!\,3!} = \boxed{56}$

(c) Contains at least 3 Heads?

$2^8 - (^8C_0 + {^8C_1} + {^8C_2}) = \boxed{219}$ | OR: $^8C_3 + {^8C_4} + {^8C_5} + {^8C_6} + {^8C_7} + {^8C_8}$

Q5: How many permutations of letter "ABCDEFGH" contains:

(a) the string ABC.

ABC _ _ _ _ _

$= 6! = \boxed{720}$

(b) the string ED

ED _ _ _ _ _ _ $= 7!$

(c) the string AB, DE, GH?

AB DE GH C F $= 5! = \boxed{120}$

(d) the string CAB, BED?

F G H CAB BED $= 4! = \boxed{24}$

considered single character as 'B' cannot be double.

(a)
_ ABC _ _ _ _
_ _ ABC _ _ _
_ _ _ ABC _ _
_ _ _ _ ABC _
_ _ _ _ _ ABC
ABC _ _ _ _ _

6! as ABC (single unit) can move to six locations.

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Q6: Suppose that a department contains 10 men & 15 women. How many ways are there to form a committee with six member if it must have:

(a) the same no. of men & women?

$6 \rightarrow$ 3 men, 3 women

$^{10}C_3 \cdot {}^{15}C_3$

(b) more women than men?

$\boxed{\text{Case 1:}}$ w=4, m=2, $\boxed{\text{Case 2}}$: w=5, m=1, $\boxed{\text{Case 3}}$: w=6, m=0.

$\left({}^{10}C_4 \cdot {}^{15}C_2\right) + \left({}^{10}C_5 \cdot {}^{15}C_2\right) + \left({}^{10}C_6 \cdot {}^{15}C_0\right)$

$\boxed{\text{Ex: 6.5}}$: Generalized Permutations & Combns:

- In many counting problems elements may be used repeatedly:

→ Permutation: A letter / digit many be used more than once on a liscence plate.

→ Combination: A specific type of fruit can be chosen repeadetly from 'n' types of fruits.

Ex: How many strings of length 8 can be formed from the upper case letter of Eng. alphabets.

Permutations with repeatation $= n^r$

$= (26)^8$

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★ IF not written, then repetation is allowed.

★ Jitne cheezon ko choose krna hota vo 'r' hoti.

★ out of jitne cheezon ko choose krna vo 'n' hoti.

Per / combo	WITH OR W/OUT REP.	FORMULA.
r-permutation	w/out rep	$\frac{n!}{(n-r)!}$
r-combi	w/out rep	$\frac{n!}{(n-r)!\,r!}$
r-permut.	with rep	n^r
r-combi.	with rep	$\frac{(n+r-1)!}{r!\,(n-1)!}$

Q: How many ways are there to select four pieces of fruit from a bowl containing apples, oranges, pears if the order in which the peices are selected does not matter, only the type of fruit and not the individual peice matters, and there are 4 peices of each fruit in the bowl?

r: jitne cheezon ko choose krna hota vo r.

combination

Apples	Oranges	Pears.
1 4A,	6 4O	11 4P
2 3A, 1O	7 3O, 1A	12 3P, 1A
3 3A, 1P	8 3O, 1P	13 3P, 1O
4 2A, 2O	9 2A, 2P	14 2O, 2P
5 2A, 1O, 1P	10 2O, 1A, 1P	15 2P, 1A, 1O

Case ↓

$\boxed{\text{= r-Combi with rep}} = \frac{(n+r-1)!}{r!\,(n-1)!} = \frac{(3+4-1)!}{4!\,(3-1)!} = \boxed{15}$

manual & formula both used.

Use ${}^nC_r$

☐ Permutation with Indistinguishable Objects.

Q: How many diff. string can be made by re-ordering the letters of word SUCCESS.

$S = 3,\quad C = 2,\quad E = 1,\quad U = 1.$

$7-3=4 \qquad 4-2=2$

$$({}^7C_3)({}^4C_2)({}^2C_1)({}^1C_1) = \frac{7!}{3!\,2!\,1!\,1!}$$

'S' placed   'C' placed   'E' placed   'U' placed

S C E U

Theorum #3: The no. of ~~[illegible]~~ permutations of n-objects where there are n_1 indistinguishable objects of type 1, n_2 indisting. of type 2, ... n_k of n_k indistinguishable of type k.; then

$$\frac{n!}{n_1!\,n_2!\cdots n_k!}$$

LECTURE - 19

Wednesday 29/10/25

Q: How many diff. ways can five elements be selected in order from a set of five elements when repetation is allowed?

$n^r = (5)^5$ Perm. w rep.

Q: How many diff. ways are there to choose a dozen donuts from the 21 varieties at a donut shop? Combination with rep.

$$\frac{(n+r-1)!}{r!(n-1)!} \rightarrow \frac{(21+12-1)!}{(21-1)!\,12!}$$

Q: A croissant shop has six varieties of croissants. How many ways are there to choose:

(a) a dozen croissants $\rightarrow \frac{(6+12-1)!}{(6-1)!\,12!}$

(b) three dozen croissants

$$\frac{(6+36-1)!}{(6-1)!\,36!}$$

Q: How many diff. strings can be made from the letter in:

(a) MISSISSIPPI ⇒ Permu. w Indistinguisable object

(b) ABRACADABRA

(a) $\frac{11!}{4!\,4!\times 2!\times 1!} = 34650.$ $M=1,\ i=4,\ s=4,\ p=2$

(b) $\frac{11!}{5!\,2!\,2!\,1!\,1!} =$ $A=5,\ B=2,\ R=2,\ C=1,\ D=1$

Q: How many diff strings can be made from the letters in AARDVARK, using all letters, if all A's must be consecutive.

$AAA = 1,\ R=2,\ V=1,\ K=1$ $n=6.$

$$= \frac{6!}{2!\,1!\,1!\,1!} = \boxed{360}$$

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Ex: 6.4 "Binomial Expansion".

$$(x+y)^n = \sum_{j=0}^{n} \binom{n}{j} x^{n-j} y^j$$

Q: $(x+y)^4 \rightarrow n=4$

$$\sum_{k=0}^{4} ({}^4C_k)(x)^{4-k}(y)^k$$

$$= {}^4C_0\, x^4 y^0 + {}^4C_1\, x^3 y^1 + {}^4C_2\, x^2 y^2 + {}^4C_3\, x^1 y^3 + {}^4C_4\, x^0 y^4$$

$$(x+y)^4 = (1)(4)(1) + 4x^3 y + 6x^2 y^2 + 4xy^3 + (1)(1)(y).$$

Q: $(2x-5y)^3 \rightarrow n=3$

$$= {}^3C_0 (2x)^3(-5y)^0 + {}^3C_1(2x)^2(-5y)^1 + {}^3C_2(2x)^1(-5y)^2 + {}^3C_3(2x)^0(-5y)^3$$

Q: What is the cofficient of $x^{12}y^{13}$ in the expansion of $(x+y)^{25}$?

$${}^{25}C_{13}\,(x)^{25-13}(y)^{13}$$

k value which tells exact term

$${}^{25}C_{13} = 5200300$$

Q: Cofficient of $x^{12}y^{13}$ in the exp of $(2x-3y)^{25}$?

$${}^{25}C_{13}\,(2x)^{12}(-3y)^{13}.$$

Cofficient $= 5200300 \cdot (2)^{12} \cdot (-3)^{13}$

$$= -3.39 \times 10^{16}.$$

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Q: Use Binomial to find the coff. of $x^a y^b$ in expansion of $(5x^2+2y^3)^6$.

(a) $a=6, \quad b=9.$

$\rightarrow x^6 y^9$ cofficient.

$$\sum_{k=0}^{6} {}^6C_k (5x^2)^{6-k} (2y^3)^k. \longrightarrow (A)$$

★ power of 'y' is 'k' ∴ $3k=9$

$$\boxed{k=3}$$

~~★ power of 'x' is '6-k' ∴ 12-2k = 6~~ $\cancel{\boxed{k=3}}$

★ Now put '$k=3$' into 'x' term of eq (A) to verify the value $a=6$ in question (to see if that term does exist)

x term $\rightarrow (5x^2)^{6-k}$

$$x^{12-2k} = x^a$$

$$12-2(3) = \boxed{6}$$ ✓

$$\therefore x^{12-2k} = x^a = x^6$$ ✓

$$\text{Coff.} = ({}^6C_3)\cdot(5)^{6-3}\cdot(2)^3 = \underline{\qquad}$$

(b) $a=3, \quad b=12$

$3k = 12$

$$\boxed{k=4}$$

★ $k=4$ in $2(6-k) \Rightarrow a=4$

since a should've been '3' (but it's 4)

∴ the $\boxed{\text{cofficient} = 0}$ as term doesn't exist.

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Q: How many Bit strings of length 10 have: (either 0 or 1)
(a) exactly three 0s ?

Advance Counting | LECTURE-20 | Thursday 6/11/25.

8.1 "Applications of Recurrence Relations".

Many counting problems cannot be solved easily using the methods in chapter 6.

Example.
Q: How many bits strings of length n do not contains two consecutive 0s ?

⇒ Modeling with Recurrence Relation (R.R)
→ Counting rabbits on Island
→ No. of moves needed in Tower of Hanoi
→ Counting bit strings with certain properties.

⇒ Recurrence Relations:
A recurrence relation for the seq. $\{a_n\}$ is an equation that expresses 'a_n' in terms of one or more of the previous terms of the seq. namely; $a_0, a_1, \dots a_{n-1}$ for all integers n with $n \geq n_0$ where $n_0 > 0$

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★ R.R is in terms of previous term (a_{n-1})
★ Closed form is in general n^{th} term and a_0.

→ The sequence is called a solution of a R.R if its terms satisfy the R.R.

→ The initial conditional for a sequence specify the terms that precede the first term where R.R takes effect.

Q: $\{a_n\} \rightarrow$ 5, 8, 11, 14, 17 ..... (a_0, a_1, a_2, a_3, a_4) — general formula for n^{th} term.
Find R.R and generate its closed form (explicit formula).

current term = (previous term) + 3
$a_n = (a_{n-1}) + 3 \quad n \geq 1$

$a_1 = a_0 + 3 \rightarrow 5 + 3 = 5 + 1(3) \quad n=1$
$a_2 = a_1 + 3 \rightarrow (5+3) + 3 = 5 + 2(3) \quad n=2$
$a_3 = a_2 + 3 \rightarrow (5+3+3) + 3 = 5 + 3(3) \quad n=3$
for closed formula

$\boxed{\text{Closed Formula}}$: $a_n = a_0 + 3n$

Q: $a_n = 2 + 5n$. Find R.R from this closed form.

$a_0 = 2 + 5(0) = 2$
$a_1 = 2 + 5(1) = 7$
$a_2 = 2 + 5(2) = 12$
$a_3 = 2 + 5(3) = 17$

R.R → $\boxed{a_n = a_{n-1} + 5}$

sequence = 2, 7, 12, 17 ....

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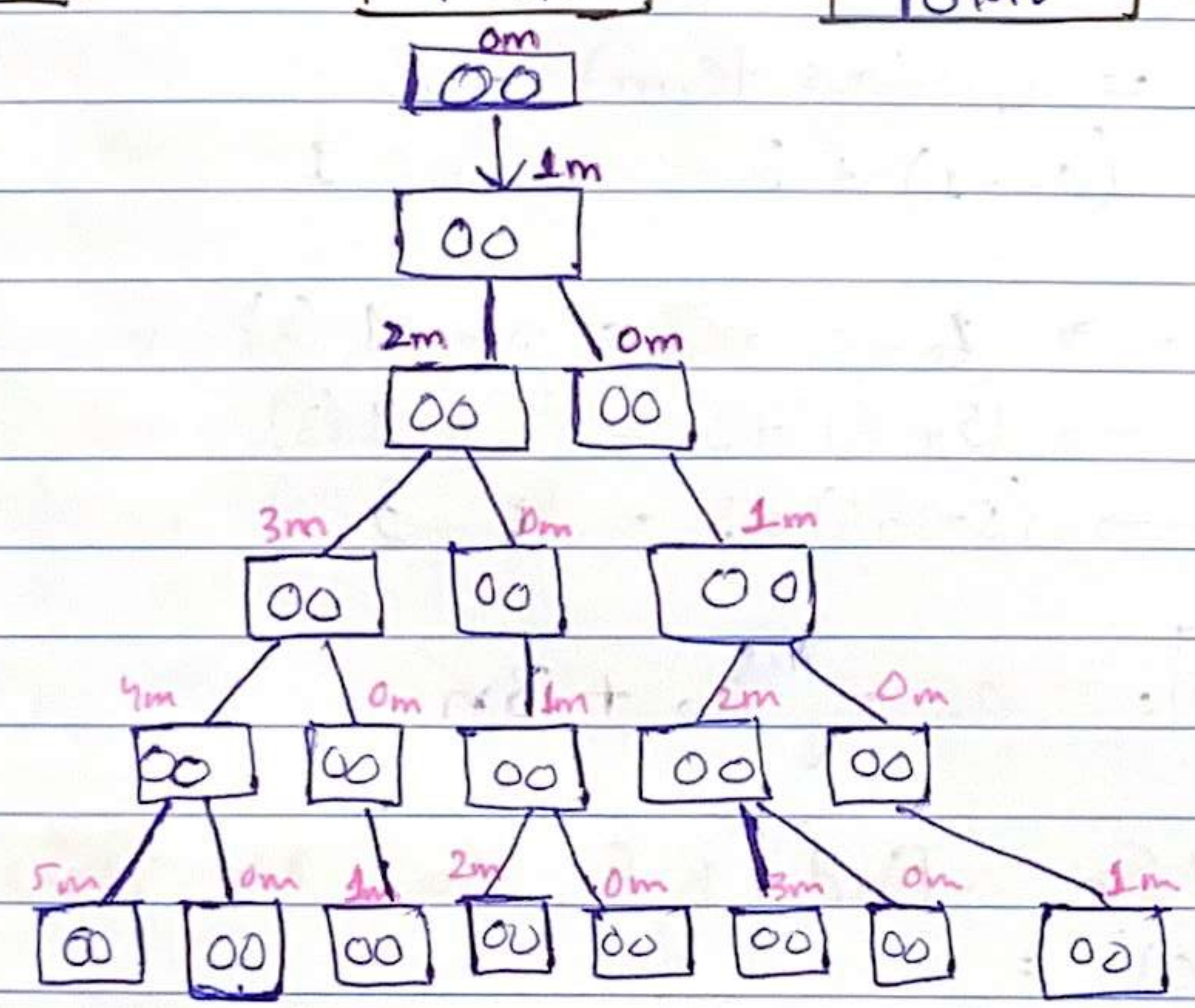
Q: A young pair of rabbits (of opposite gender) is placed on an island. A pair of rabbits does not breed until they are 2 months old. After being 2 months old, each pair of rabbits produces another pair each month.
Find RR of the no. of pairs of rabbits on the island after 'n' months, assuming that no rabbits ever die.



Months	Total
1	1
2	1
3	2
4	3
5	5
6	8

$\boxed{\text{Sequence:}}$ 1, 1, 2, 3, 5, 8 . . . . Fabionacci

$\boxed{R \cdot R}$: $a_n = a_{n-1} + a_{n-2}$ $n \geq 3$ $\begin{cases} a_1 = 1 \\ a_2 = 1 \end{cases}$

initial condition: fixed.

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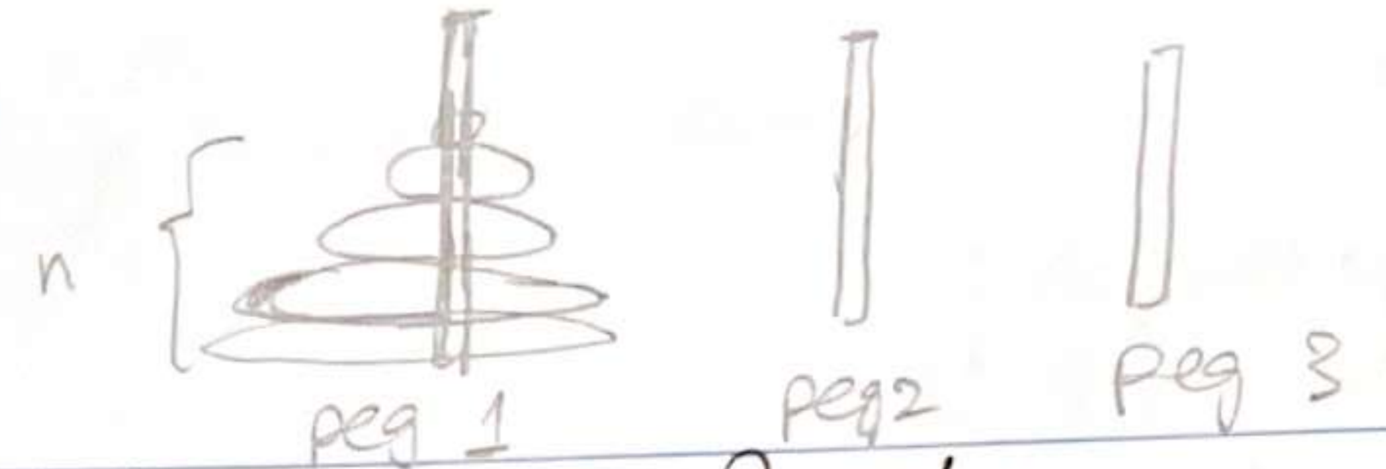

The tower of Hanoi Puzzle:

Q: puzzle consists of 3 pegs mounted on a board together with disks of diff. sizes.
Initially these disks are placed on the first peg in order of size, with largest on the top.
The rules of puzzle allow disks to be moved one at a time from peg 1 to peg 2. as long as a disk is never placed on the top of a smaller disk.
The goal of the puzzle is to have all the disks on the second peg in order of size, having largest disk on bottom.
Let H_n denote the no. of moves needed to solve the tower of Hanoi puzzle with 'n' disks
→ Find R.R for $\{H_n\}$.

* Since peg 3 does not have any condition; so we can place any no. of disk in peg 3 without any order of size

$\boxed{\text{Step 1:}}$ move $n-1$ disk from peg 1 to peg 3.

$\boxed{\text{Step 2}}$: move the last disk (largest) in peg 1 to peg 2

$\boxed{\text{Step 3}}$: move $n-2$ from peg 3 to peg 1 (which is empty rn). which leaves one disk (2^{nd} largest) in peg 3.

$\boxed{\text{Step 4:}}$ move the only disk (2^{nd} largest) from peg 3 → peg 2.

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★ After step 4; peg 2 have 2nd largest on top of largest.

★ Moving n-1 disks is being done twice in step 1 and 3 whereas; moving largest disk to peg 2 is done only 1 time

$$\boxed{H_n = 2(H_{n-1}) + 1}$$

$= 2(2H_{n-2}+1) + 1 \longrightarrow 2^2 H_{n-2} + 2 + 1$

$= 2(2(2H_{n-2}+1)+1) + 1 \longrightarrow 2^3 H_{n-3} + 2^2 + 2 + 1$

$\vdots$

$= 2^{n-1} H_1 + 2^{n-2} + 2^{n-3} + \cdots 2^2 + 2^1 + 1$

$= 2^{n-1} + 2^{n-2} + 2^{n-3} + \cdots + 2^2 + 2 + 1$

LECTURE-21

Monday 10/11/25.

Q: Find a recurrence relation and give initial conditions for the number of bit strings for length 'n' that do not have two consecutive zeros. How many such bit strings are there of length five?

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cancelling consecutive zeros.

	ENDING WITH 0	1	Combinations at specific 'n' after cutting consec. zeros term.
n = 1	0	1	2
n=2	~~00~~ 10	01 11	3
n=3	~~000~~ ~~100~~ 010 110	011 ~~001~~ 101 111	5
n=4	~~0000~~ ~~1000~~ ~~0100~~ ~~0010~~ 0110 ~~1100~~ 1110 1010	0101 ~~0001~~ ~~1001~~ ~~0011~~ 1011 0111 1111	8

⇒ENDS at zero: ... _ _ 1 0

- For ending at zero; we can observe from table above that last "two" positions are fixed with '10' ∴ we can have "n-2" combinations before last '10' in blanks.

⇒ENDS at one: ... _ _ _ 1

- For ending at one; observe from table that last element 1 is fixed ∴ Blanks can have "n-1" combinations.